INDEPENDENT PORTFOLIO PROJECT | SYNTHETIC DATA

Civic Operating Budget Analysis

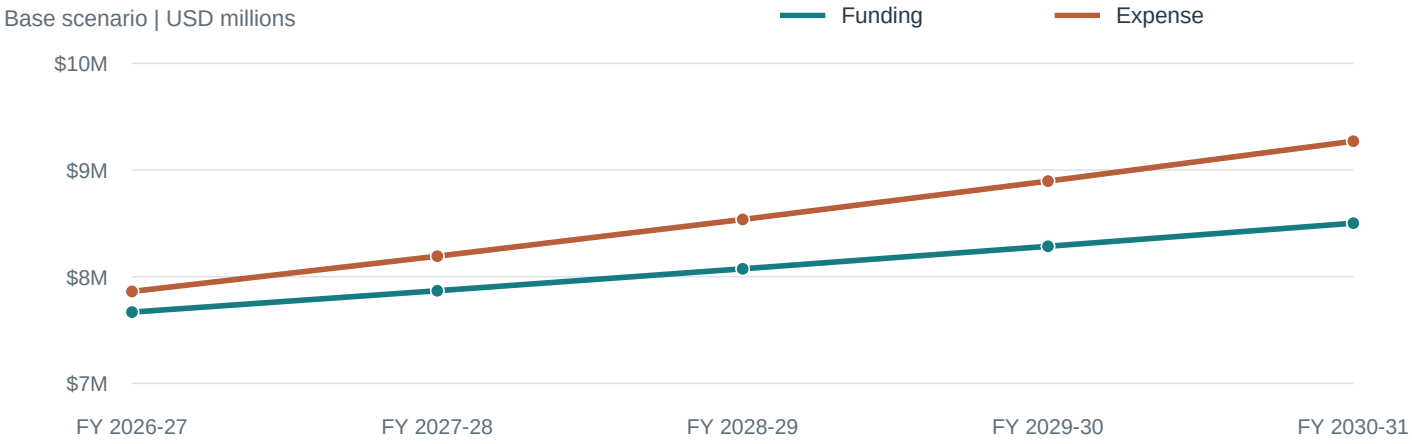
Excel reporting, SQL analysis and scenario-based operating planning.

Sileshi Hirpa | October 23, 2026 | Base scenario

$325,050	$16.73M	$786,463
FY 2025-26 expense above budget	FY 2027-29 proposed expense	FY 2027-29 modeled funding gap

Purpose

Translate monthly financial records into a clear budget review: identify expense variances, compare departments, project funding and expense over five years, and summarize a two-year spending proposal.



Three findings for a finance reviewer

Expense review: Public Works has the largest synthetic department overspend, led by personnel. Account-level review helps focus follow-up questions.

Planning outlook: Assumed expense growth exceeds funding growth. The modeled annual shortfall widens from $193,747 to $768,671 over the forecast horizon.

Two-year proposal: The FY 2027-29 proposal includes FY 2027-28 and FY 2028-29. It is a planning estimate, not an adopted appropriation.

Scope: 360 monthly records, three fictional departments and five accounts. No City of Alameda financial records, employment relationship or ERP experience is represented.

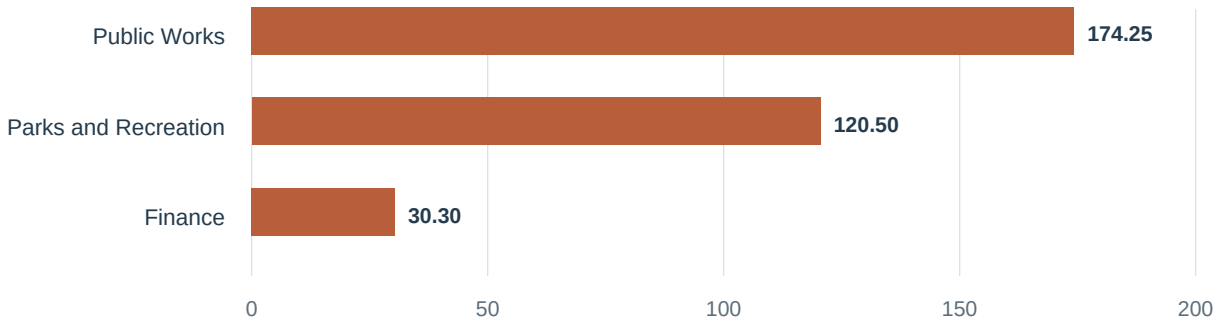
01 | Budget performance

Where spending differs from the plan

Full FY 2025-26. Favorable expense variance equals budget minus actual. A negative value means expense exceeded budget.

Department	Budget	Actual	Favorable variance
Public Works	$3,650,000	$3,824,250	($174,250)
Parks and Recreation	$2,600,000	$2,720,500	($120,500)
Finance	$970,000	$1,000,300	($30,300)

Expense above budget | USD thousands



Largest account exceptions

Department / account	Overspend	Above budget
Public Works / Personnel	$132,000	6.0%
Parks and Recreation / Materials	$56,000	16.0%
Parks and Recreation / Personnel	$42,500	2.5%
Public Works / Contract services	$26,250	2.5%

How to use this review

Confirm the accounting period and classification first. Then investigate whether each variance reflects timing, a one-time item, a recurring price change, or a service-volume change. Obtain operational explanations before proposing a budget adjustment.

The synthetic differences were deliberately generated. They demonstrate an exception-review process and do not establish actual departmental inefficiency, staffing needs or achievable savings.

Source: sql/queries/department_variance.sql and account_exceptions.sql; reports/exports contains the corresponding CSV outputs.

02 | Operating outlook

Five years of projections. One two-year proposal.

The model compounds each department/account baseline using editable annual growth assumptions. Dollar amounts are rounded to cents before projecting the next year.

Fiscal year	Funding	Operating expense	Funding less expense
2026-27	$7,668,233	$7,861,980	($193,747)
2027-28	$7,868,598	$8,192,400	($323,802)
2028-29	$8,074,233	$8,536,893	($462,660)
2029-30	$8,285,275	$8,896,063	($610,787)
2030-31	$8,501,870	$9,270,541	($768,671)

FY 2027-29 proposal | Base scenario

Measure	Two-year amount
Assigned support + service fees	$15,942,830.54
Operating expense	$16,729,293.39
Funding less expense	($786,462.85)

Growth assumptions | Annual rates

Driver	Base	Constrained
Assigned operating support	2.5%	0.5%
Service fees	3.0%	1.0%
Contract services	4.0%	6.0%
Materials	3.0%	5.0%
Personnel	4.5%	5.5%

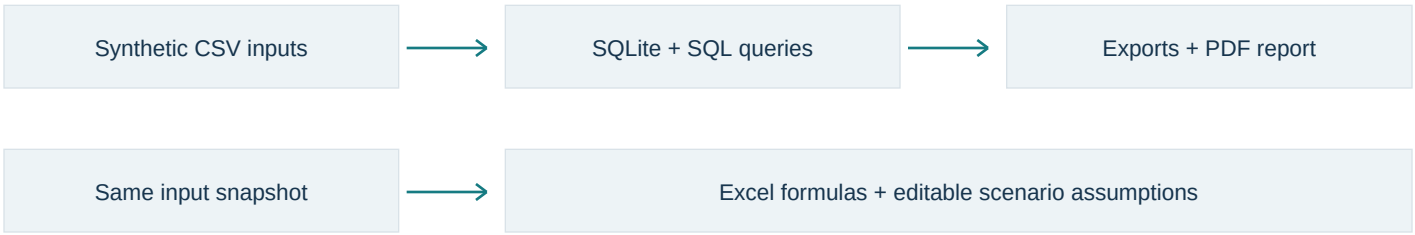
The same illustrative rates apply in each of the five forecast years in the published snapshot. Constrained assumptions are provided for workbook exploration; the reported amounts above use Base. These are judgmental inputs, not statistically estimated predictions.

Assigned support is funding allocated to this three-department scope. It is not additional consolidated municipal revenue. A funding gap is not a cash, reserve or fund-balance calculation.

Source: sql/queries/five_year_forecast.sql and data/raw/forecast_rates.csv. FY labels use a July-June convention and the ending year.

03 | Methods and confidence

Traceable calculations. Clear practical limits.



Methods used

Area	Implementation
Data integrity	Composite monthly keys, reference-table joins, fiscal-period checks and validated integer-cent amounts.
SQL analysis	Grouped views, conditional variance signs, DENSE_RANK and ROW_NUMBER for exception review, and a recursive forecast CTE.
Excel planning	SUMIFS and COUNTIFS, one scenario selector, linked growth assumptions, formula-based forecasts and a native chart.
Reconciliation	Compare source controls, department totals and saved forecast values against independent SQL results.

Validation evidence

Nine model tests passed. Tests cover duplicates, missing records/drivers, foreign keys, fiscal cutoffs, zero denominators, variance direction and the -100% growth boundary.

150 SQL forecast results checked: 75 account/year amounts in each of two scenarios matched independent Decimal calculations. All 75 saved Base workbook forecast amounts also matched SQL. Historical actuals stayed unchanged in workbook scenario checks.

Refresh and verification limits

Excel contains an embedded snapshot. SQL rebuilds do not automatically refresh Excel, and edits in Excel do not update SQLite. Desktop Excel recalculation was not independently certified. The forecast is an operating model, without detailed position control, capital planning, fund accounting, cash flow or reserve schedules.

Authorship and evidence

Prepared by Sileshi Hirpa with AI assistance, drawing on Excel and SQL coursework concepts. Synthetic data and assumptions are disclosed throughout. The project demonstrates analytical work and its verification; it claims no actual municipal savings, employer outcomes or ERP implementation.

Review the workbook in excel/, SQL in sql/, the methodology in docs/, and the test and validation evidence in reports/validation/. The repository provides the underlying files for inspection and reproduction.